Measuring the Elementary Charge with Millikan's Oil-Drop Method and Avogadro's Number

PY2107 – Experiment 6

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1 Abstract

Millikan's 1913 oil-drop experiment was reproduced to test whether electric charge is quantised and to measure the elementary charge e. The terminal falling and rising speeds of 28 charged oil drops in air, with and without an applied electric field, were used to determine each drop's radius and charge. As in Millikan's original work, the charges did not form a continuum but fell into three discrete levels, $n = 1, 2, 3$, in the ratio $1 : 1.94 : 3.05$, close to $1 : 2 : 3$. Millikan's central step, the slip (Cunningham) correction for drops comparable in size to the mean free path of air, was applied with $A = 1.188$, calculated from the Leybold slip constant and a molecular diameter of 0.370 nm. This gives $e = (1.7 \pm 0.1) \times 10^{-19}\,\mathrm{C}$, in agreement with the accepted value of $1.602 \times 10^{-19}\,\mathrm{C}$ and with Millikan's own result of approximately $1.59 \times 10^{-19}\,\mathrm{C}$. Fitting A to the data, as Millikan did, gives $A = 0.69 \pm 0.55$, consistent with the analytic value within 0.9σ but too weakly constrained to test it. This fitted A implies $N_\mathrm{A} = (3.5 \pm 2.8) \times 10^{23}\,\mathrm{mol}^{-1}$, consistent with the accepted value.

2 Theory

An oil drop falling through air experiences gravity and viscous drag. As in the lab briefing, buoyancy is neglected because the oil density is much greater than the air density ($\rho_{oil} \gg \rho_{air}$). The gravitational force is therefore

$$F_g = \frac{4}{3}\pi r^3 \rho g, \tag{1}$$

where r is the drop radius and ρ is the oil density. For a small spherical drop, Stokes' law is corrected for non-continuum flow using the Cunningham correction factor, following Cunningham's analysis of small spherical particles moving through gases [1]. A first order correction term is used as detailed in Appendix A.1.

$$C_{\mathrm{c}}(r,p) = 1 + \frac{b}{rp} = 1 + \frac{A}{r}, \qquad A = \frac{b}{p}, \tag{2}$$

where p is the air pressure, $b = 80\,\mu\mathrm{m\,hPa}$ is the constant specified for the Leybold apparatus, and A is the pressure-dependent slip length. Oxford independently reports $b = 7.88 \times 10^{-3}\,\mathrm{Pa \cdot m}$, consistent with the Leybold value $8.0 \times 10^{-3}\,\mathrm{Pa \cdot m}$ after unit conversion [6]. The corrected drag force is therefore

$$F_D = \frac{6\pi\eta r v}{C_{\mathrm{c}}(r,p)}. \tag{3}$$

Here η is the viscosity of air and v is the drop speed. The pressure-dependent correction and its constant are given in the Leybold apparatus documentation [3, 6]. With the electric field off, a drop falling at terminal speed v_g satisfies

$$\frac{6\pi\eta r v_g}{C_{\mathrm{c}}(r,p)} = \frac{4}{3}\pi r^3 \rho g, \tag{4}$$

giving us,

$$r = \sqrt{\frac{9\eta v_g}{2\rho g}}. \tag{5}$$

When a potential difference V is applied across plates separated by distance d, the electric field is approximately $E = V/d$. The electric force on a drop of charge q is $F_E = qE$. If the drop rises at terminal speed v_E, the electric force balances the drop's weight and the drag opposing the upward motion. Combining the terminal-fall relation with the charge force-balance gives the uncorrected charge q_0 and the Leybold slip-corrected charge [3]:

$$q_0 = \frac{4\pi d}{3V}\left(\frac{(9\eta/2)^3}{\rho g}\right)^{1/2}(v_g + v_E)\sqrt{v_g}, \qquad q = \frac{q_0}{C_{\mathrm{c}}(r,p)^{3/2}} = \frac{q_0}{(1 + A/r)^{3/2}}. \tag{6}$$

For a stationary drop, $v_E = 0$. Millikan's observation is that measured drop charges occur in integer multiples of the elementary charge:

$$q = ne, \tag{7}$$

where n is an integer and e is the elementary charge, as observed by Millikan [4]. A fit of q against n therefore estimates e from its slope.

When Millikan originally performed this experiment Cunningham's slip correction factor had been proposed 3 year earlier [1]. Millikan was the first to measure its value. His method involved plotting his apparent charge result before correction against the inverse radius as follow,

$$q_{app}^{2/3} = q_1^{2/3}\left(q_1^{2/3} \cdot \frac{A\lambda}{r}\right) \tag{8}$$

3 Experimental Procedure

A fine spray of oil was introduced into the region between the parallel plates of the Leybold oil-drop apparatus. The plate separation was $d = 6\,\text{mm}$, and each major graticule division represented 0.5 mm. Individual drops were observed through the microscope, and a suitable drop was selected and followed as it moved through the field of view. With the electric field switched off, the time taken to travel a known distance was measured. This terminal falling speed, v_g, was used with Equation 4 to determine the drop radius, including the Cunningham correction for slip in the non-continuum regime.

A potential difference was then applied across the plates. The voltage and the drop's motion were recorded, and its terminal rising speed v_E (or zero for a stationary drop) was determined over the measured distance. The falling and rising speeds, plate separation, applied voltage, and calculated radius were used in Equation 6 to calculate the drop's charge. This process was repeated for a number of drops. The resulting charges were grouped by level, with each level corresponding to an integer multiple n of the elementary charge, and the charge-versus-level data were analysed to estimate e.



Figure 1: Leybold apparatus as used in this experiment

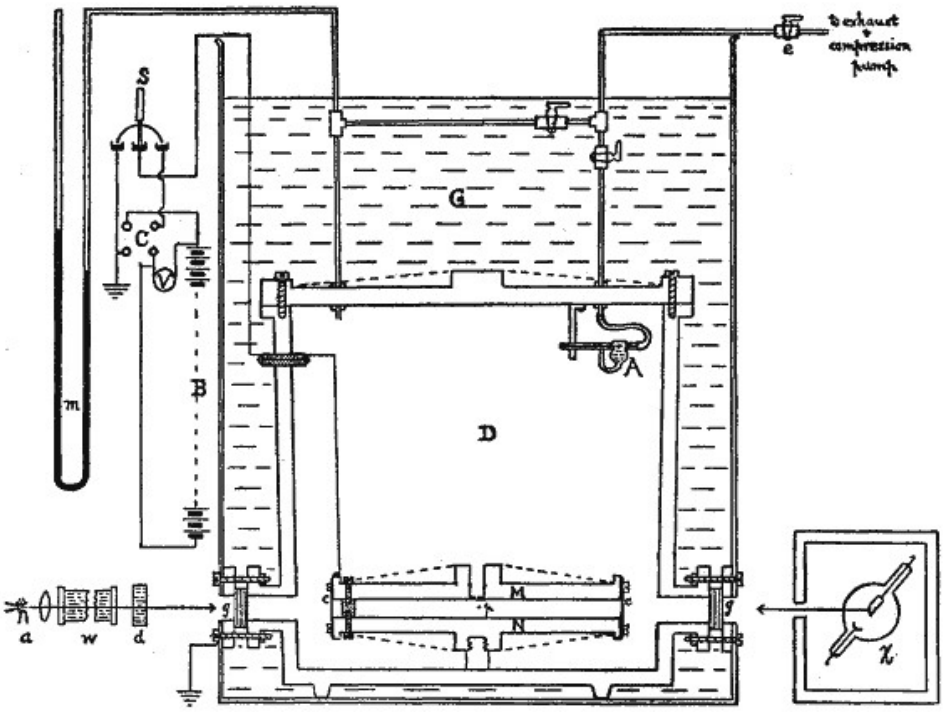


Figure 2: Diagram of the original apparatus as used by Millikan in 1913.

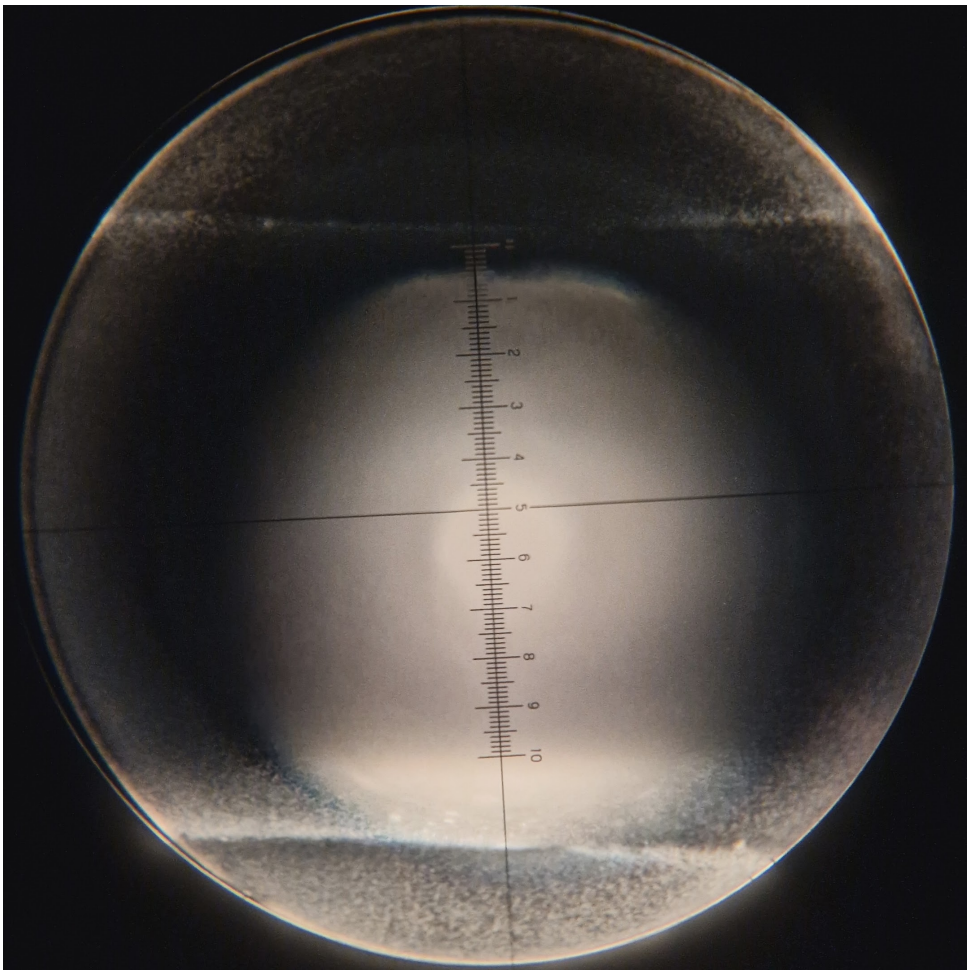


Figure 3: Viewfinder of the Leybold oil-drop apparatus. Contrast boosted. Drops visible on R.H.S.

4 Results

Table 1: Summary of experimental parameters.

Parameter	Symbol	Value
Plate separation (treated as exact)	d	$6\,\mathrm{mm}$
Graticule division	L	$0.5\,\mathrm{mm}$
Temperature	T	$(23.4 \pm 0.1)\,\mathrm{C}$
Air viscosity (treated as exact)	η	$1.83 \times 10^{-5}\,\mathrm{Pa \cdot s}$
Oil density (treated as exact)	ρ	$900\,\mathrm{kg \cdot m^{-3}}$
Slip constant (treated as exact)	b	$8.0 \times 10^{-3}\,\mathrm{Pa \cdot m}$
Atmospheric pressure	P	$(1000 \pm 4)\,\mathrm{hPa}$

Table 2: Recomputed drop values with propagated one-standard-uncertainty errors. $v_E = 0$ denotes a stationary drop. The first column gives the assigned charge level n.

Drop / n	v_g ($10^{-5}\,\mathrm{m \cdot s^{-1}}$)	r ($10^{-7}\,\mathrm{m}$)	V (V)	v_E ($10^{-5}\,\mathrm{m \cdot s^{-1}}$)	q ($10^{-19}\,\mathrm{C}$)	q/n ($10^{-19}\,\mathrm{C}$)
1 / 1	5.282 ± 0.880	6.630 ± 0.584	573 ± 0.5	2.925 ± 0.488	1.754 ± 0.374	1.754 ± 0.374
2 / 1	5.474 ± 0.912	6.757 ± 0.595	568 ± 0.5	2.783 ± 0.464	1.818 ± 0.391	1.818 ± 0.391
3 / 3	11.565 ± 1.928	9.994 ± 0.865	418 ± 0.5	0	5.299 ± 1.376	1.766 ± 0.459
4 / 1	4.820 ± 0.803	6.317 ± 0.558	327 ± 0.5	0	1.711 ± 0.453	1.711 ± 0.453
5 / 1	4.820 ± 0.803	6.317 ± 0.558	565 ± 0.5	3.833 ± 0.958	1.777 ± 0.392	1.777 ± 0.392
6 / 1	5.144 ± 0.857	6.538 ± 0.576	345 ± 0.5	0	1.798 ± 0.475	1.798 ± 0.475
7 / 1	5.144 ± 0.857	6.538 ± 0.576	572 ± 0.5	3.461 ± 0.577	1.814 ± 0.378	1.814 ± 0.378
8 / 2	5.334 ± 0.889	6.665 ± 0.587	193 ± 0.5	0	3.404 ± 0.899	1.702 ± 0.450
9 / 3	8.170 ± 1.362	8.339 ± 0.727	233 ± 0.5	0	5.522 ± 1.444	1.841 ± 0.481
10 / 3	8.170 ± 1.362	8.339 ± 0.727	578 ± 0.5	11.287 ± 1.881	5.301 ± 1.013	1.767 ± 0.338
11 / 3	8.170 ± 1.362	8.339 ± 0.727	350 ± 0.5	3.865 ± 0.966	5.415 ± 1.207	1.805 ± 0.402
12 / 1	5.369 ± 0.895	6.688 ± 0.589	578 ± 0.5	2.485 ± 0.621	1.680 ± 0.379	1.680 ± 0.379
13 / 1	6.053 ± 1.009	7.125 ± 0.625	460 ± 0.5	0	1.745 ± 0.459	1.745 ± 0.459
14 / 1	6.053 ± 1.009	7.125 ± 0.625	578 ± 0.5	1.749 ± 0.875	1.790 ± 0.451	1.790 ± 0.451
15 / 1	5.503 ± 0.917	6.775 ± 0.596	578 ± 0.5	2.690 ± 0.673	1.778 ± 0.400	1.778 ± 0.400
16 / 1	5.710 ± 0.952	6.909 ± 0.607	407 ± 0.5	0	1.798 ± 0.474	1.798 ± 0.474
17 / 3	10.115 ± 1.686	9.321 ± 0.809	333 ± 0.5	0	5.397 ± 1.405	1.799 ± 0.468
18 / 3	10.115 ± 1.686	9.321 ± 0.809	578 ± 0.5	7.393 ± 1.232	5.382 ± 1.090	1.794 ± 0.363
19 / 2	8.586 ± 1.431	8.558 ± 0.745	578 ± 0.5	3.372 ± 0.562	3.351 ± 0.735	1.676 ± 0.367
20 / 2	8.586 ± 1.431	8.558 ± 0.745	382 ± 0.5	0	3.641 ± 0.951	1.820 ± 0.475
21 / 1	5.803 ± 0.967	6.968 ± 0.612	438 ± 0.5	0	1.714 ± 0.452	1.714 ± 0.452
22 / 1	5.803 ± 0.967	6.968 ± 0.612	578 ± 0.5	2.175 ± 0.544	1.785 ± 0.408	1.785 ± 0.408
23 / 1	4.088 ± 0.681	5.788 ± 0.514	246 ± 0.5	0	1.749 ± 0.466	1.749 ± 0.466
24 / 1	4.088 ± 0.681	5.788 ± 0.514	591 ± 0.5	6.017 ± 1.003	1.800 ± 0.349	1.800 ± 0.349
25 / 2	5.503 ± 0.917	6.775 ± 0.596	590 ± 0.5	10.183 ± 1.697	3.334 ± 0.632	1.667 ± 0.316
26 / 2	5.503 ± 0.917	6.775 ± 0.596	208 ± 0.5	0	3.318 ± 0.876	1.659 ± 0.438
27 / 2	5.938 ± 0.990	7.053 ± 0.619	590 ± 0.5	10.183 ± 1.697	3.582 ± 0.681	1.791 ± 0.340
28 / 2	5.938 ± 0.990	7.053 ± 0.619	227 ± 0.5	0	3.430 ± 0.903	1.715 ± 0.452

5 Analysis

The charges were grouped into three observed levels and assigned integer indices $n = 1, 2, 3$, without using the accepted value of e to set the levels. The groups were identified by clustering the measured charges and ordered by increasing mean charge. The mean charge at each level was divided by its index to estimate e. The resulting estimates are summarised in Table 3.

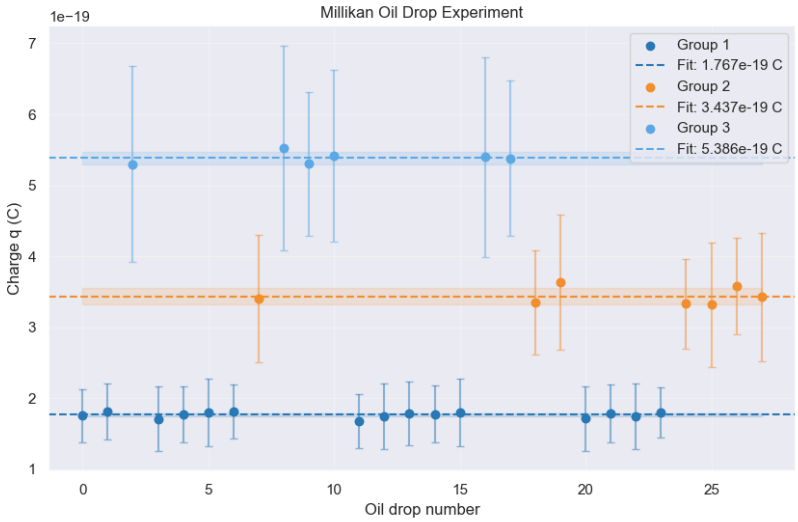


Figure 4: Measured charge against drop number. Dashed horizontal lines show the mean charge of each group; this plot illustrates the three observed charge levels and is not the charge-versus-level fit used to estimate e.

Table 3: Elementary-charge estimates obtained from each measurement level.

Level	Measurements	q (C)	$e = q/n$ (C)
1	15	$(1.7672 \pm 0.0106) \times 10^{-19}$	$(1.7672 \pm 0.0106) \times 10^{-19}$
2	7	$(3.4372 \pm 0.0478) \times 10^{-19}$	$(1.7186 \pm 0.0239) \times 10^{-19}$
3	6	$(5.3857 \pm 0.0338) \times 10^{-19}$	$(1.7952 \pm 0.0113) \times 10^{-19}$

The weighted mean of the three level estimates is

$$e_{\text{weighted}} = (1.7745 \pm 0.0073) \times 10^{-19}\,\text{C}. \tag{9}$$

Using the propagated per-drop uncertainties and the shared pressure uncertainty as a common covariance contribution, a generalised least-squares fit of charge against the assigned integer n, as seen in fig. 6, gives

$$e = (1.7590 \pm 0.0769) \times 10^{-19}\,\text{C}, \qquad \chi^2/\text{dof} = 0.02, \tag{10}$$

The fit with a free intercept gives $e = (1.7483 \pm 0.205) \times 10^{-19}$ C and an intercept of $(0.0148 \pm 0.263) \times 10^{-19}$ C. The intercept is only 0.1σ from zero, so the fit provides no evidence of a significant constant offset in the measured charges. Its reduced chi-squared is also 0.02.

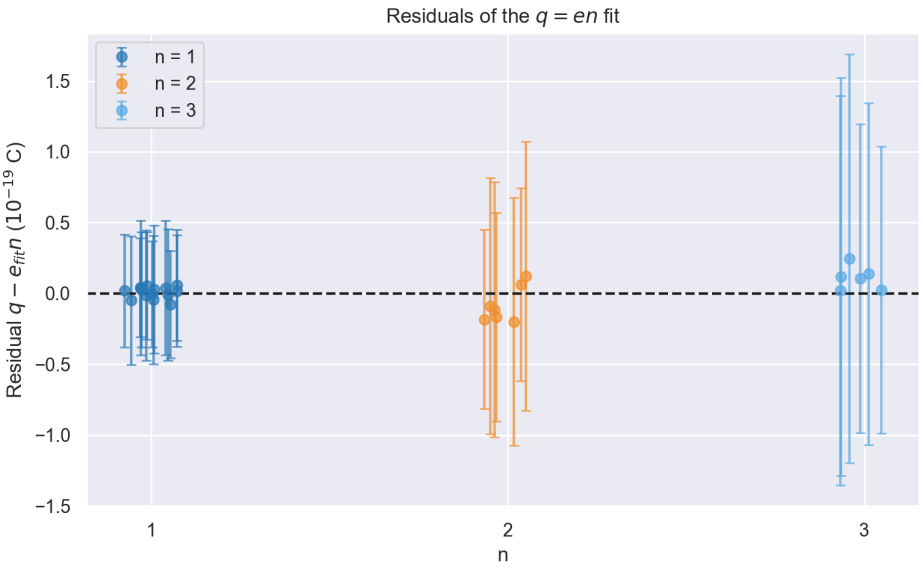


Figure 5: Residuals $q - e_{\text{fit}}n$ of the origin-constrained fit for each drop, with propagated uncertainties. The residuals scatter about zero with no systematic trend with n.

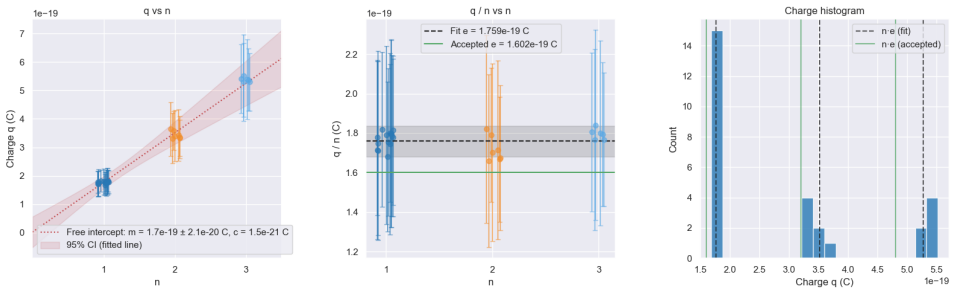


Figure 6: Combined analysis of the Millikan experiment, showing the charge histogram, q/n against n, and the residuals of the origin-constrained fit.

The level estimates have a consistency statistic of $\chi^2 = 9.34$ for two degrees of freedom. The weighted mean is 10.8% above the accepted elementary charge, $1.602176634 \times 10^{-19}$ C; the origin-constrained fit is 9.8% high. The level statistic indicates more scatter between the three level estimates than expected from their within-group standard errors. The fit uncertainties instead use the propagated reading errors and common pressure contribution, so they answer a different uncertainty question.

5.1 Apparent-charge correction and revised e_1

The apparent charge q_{app} is calculated before applying the slip correction with known A and the radius from Eq. 5. Following Millikan's method, the true charge q and the apparent charge are related by

$$q_{\text{app}}^{2/3} = q^{2/3}\left(1 + A\frac{\lambda}{r}\right) = n^{2/3}e_1^{2/3}\left(1 + A\frac{\lambda}{r}\right), \qquad q = ne_1, \tag{11}$$

so in terms of the apparent charge per unit, $e_{\text{app}} = q_{\text{app}}/n$, the drops of every charge level lie on a single straight line,

$$e_{\text{app}}^{2/3} = e_1^{2/3} + e_1^{2/3}A\frac{\lambda}{r} = c + m\frac{\lambda}{r}, \qquad c = e_1^{2/3}, \quad m = cA. \tag{12}$$

The intercept is $c = e_1^{2/3}$ and the slip constant is the slope-to-intercept ratio, $A = m/c$. The charges were assigned to levels $n = 1, 2, 3$ by clustering the apparent charges (15, 7 and 6 drops). All drops were fitted simultaneously with a single line with two free parameters, $c = e_1^{2/3}$ and $m = cA$, and the radius was recalculated self-consistently from the slip length $A\lambda$. Figure 7 shows the data, the fit and its residuals.

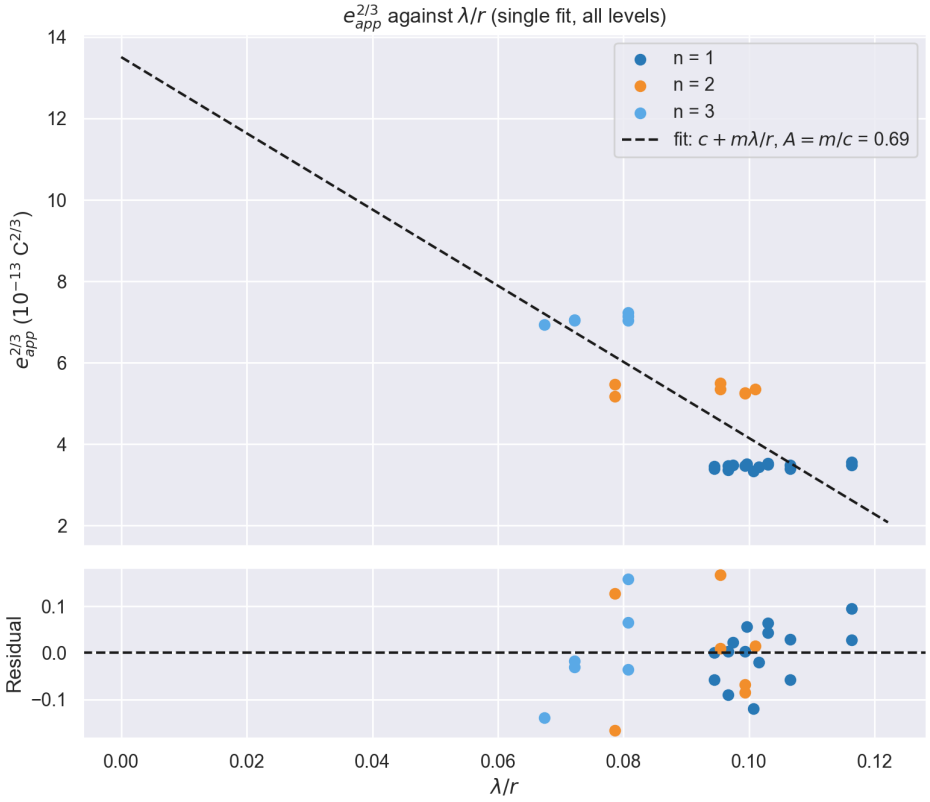


Figure 7: $e_{\text{app}}^{2/3}$ against λ/r for all charge levels, with the single fit $c + m\lambda/r$ (dashed) and its residuals (lower panel).

The fitted A is extremely weakly constrained because λ/r spans only 0.0674–0.1163. The

fit gives

$$A_{\text{fit}} = 0.6927 \pm 0.55, \qquad A_{\text{fit}}\lambda = 4.6629 \times 10^{-8}\,\text{m}. \tag{13}$$

The fitted intercept is $c = (1.3506 \times 10^{-13} \pm 2.63 \times 10^{-12})\,\text{C}^{2/3}$ and the slope is $m = (0.936 \pm 458.753) \times 10^{-13}\,\text{C}^{2/3}$. Using the reported uncertainty $u_A = 0.55$, the intercept implies $e_1 = (4.9633 \times 10^{-20} \pm 1.45 \times 10^{-18})\,\text{C}$, but the fit uncertainties are so large that this is not a useful estimate.

Analytic A. Since $A\lambda = b/p$ and $\lambda \propto 1/p$, the pressure cancels:

$$A = \frac{b}{p\lambda} = \frac{\sqrt{2}\pi d^2 b}{k_{\text{B}}T} = 1.188 \pm 0.0004, \tag{14}$$

using $b = 8.0 \times 10^{-3}\,\text{Pa m}$ (Leybold), $d = 0.370\,\text{nm}$ and $T = 296.55\,\text{K}$; the quoted uncertainty comes only from $u_T = 0.1\,\text{K}$ and excludes uncertainty in b and d. The slip length is $A\lambda = 8.0 \times 10^{-8}\,\text{m}$. The fitted value is 0.583 times the analytic one, a difference of 0.9σ when the reported fit uncertainty is used. The analytic A is therefore adopted, and the free fit is treated as a consistency check.

The apparent charges were corrected with the analytic A using

$$q_i = \frac{q_{\text{app},i}}{(1 + A\lambda/r_i)^{3/2}}. \tag{15}$$

With A fixed, the level estimates are:

Level	Measurements	e_n (10^{-19} C)
1	15	1.7148 ± 0.0103
2	7	1.6676 ± 0.0232
3	6	1.7420 ± 0.0109

An unweighted least-squares fit of $q_n = ne_1$ to all 28 drops with A fixed gives

$$e_1 = (1.72 \pm 0.01) \times 10^{-19}\,\text{C}. \tag{16}$$

The uncertainty is the statistical uncertainty from the scatter about the fit; it does not include uncertainty in A, b, the molecular diameter, or shared apparatus constants. Freeing A gives $e_1 = (4.9633 \times 10^{-20} \pm 1.45 \times 10^{-18})\,\text{C}$ (-69.0% relative to the accepted value), but the uncertainty is much larger than the estimate itself; this result is not a useful measurement.

6 Error Analysis

The falling and rising speeds were recalculated from the measured travel distance s and time t using $v = s/t$. A graticule division is 0.5 mm; the recorded integer division counts were assigned a reading uncertainty of half a division, $u_s = 0.25\,\mathrm{mm}$. Time and voltage uncertainties were taken as half the last recorded digit. Assuming independent uncertainties in distance and time, the speed uncertainty is

$$u_v^2 = \left(\frac{\partial v}{\partial s}u_s\right)^2 + \left(\frac{\partial v}{\partial t}u_t\right)^2 = \left(\frac{u_s}{t}\right)^2 + \left(\frac{s\,u_t}{t^2}\right)^2, \tag{17}$$

or, equivalently,

$$\left(\frac{u_v}{v}\right)^2 = \left(\frac{u_s}{s}\right)^2 + \left(\frac{u_t}{t}\right)^2. \tag{18}$$

This calculation was applied separately to the falling speed v_g and the rising speed v_E. A missing rising-time measurement represents a stationary drop; in that case v_E and its uncertainty were set to zero.

For each drop, the charge was calculated using

$$q = \frac{q_0}{C_\mathrm{c}^{3/2}}, \qquad C_\mathrm{c} = 1 + \frac{A}{r}, \qquad A = \frac{b}{p}, \tag{19}$$

where q_0 and the corrected radius r are given in the Theory section. The charge depends on the measured and apparatus quantities

$$\boldsymbol{x} = (d, V, \eta, \rho, v_g, v_E, p, b). \tag{20}$$

For independent input uncertainties, the propagated charge uncertainty is

$$u_q^2 = \sum_j \left(\frac{\partial q}{\partial x_j}u_{x_j}\right)^2. \tag{21}$$

The derivatives in Equation 21 include the dependence of both r and C_c on the input quantities. They were evaluated numerically using central finite differences. For this calculation, $d = 6\,\mathrm{mm}$, $\eta = 1.83 \times 10^{-5}\,\mathrm{Pa\cdot s}$, $\rho = 900\,\mathrm{kg\cdot m^{-3}}$, and $b = 8.0 \times 10^{-3}\,\mathrm{Pa\cdot m}$ were treated as exact; the nonzero apparatus uncertainty was $u_p = 4\,\mathrm{hPa}$. Thus these results are a propagated reading-and-pressure budget, not a complete budget including uncertainties in the exact-treated constants. Since pressure is shared by all measurements, the corresponding covariance terms were included in the fits:

$$u_q^2 = \sum_j \sum_k \frac{\partial q}{\partial x_j}\frac{\partial q}{\partial x_k}\,\mathrm{Cov}(x_j, x_k). \tag{22}$$

Before applying the slip correction, the uncorrected charge has the relative uncertainty

$$\begin{aligned}\left(\frac{\sigma_{q_0}}{q_0}\right)^2 = {} & \left(\frac{\sigma_d}{d}\right)^2 + \left(\frac{\sigma_V}{V}\right)^2 + \left(\frac{3\sigma_\eta}{2\eta}\right)^2 + \left(\frac{\sigma_\rho}{2\rho}\right)^2 + \left(\frac{\sigma_g}{2g}\right)^2 \\ & + \left[\left(\frac{1}{v_g + v_E} + \frac{1}{2v_g}\right)\sigma_{v_g}\right]^2 + \left(\frac{\sigma_{v_E}}{v_g + v_E}\right)^2.\end{aligned} \tag{23}$$

This expression is for q_0 only; the final uncertainty in q must also account for the Cunningham correction through the full propagation above.

For each charge level, the mean charge and its standard error were calculated from the N recomputed charges in that group:

$$\bar{q} = \frac{1}{N}\sum_{i=1}^{N} q_i, \qquad \sigma_{\bar{q},\mathrm{stat}} = \frac{s_q}{\sqrt{N}}, \qquad s_q = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(q_i - \bar{q})^2}. \tag{24}$$

For a group assigned level n, the elementary-charge estimate and its statistical uncertainty are

$$e_n = \frac{\bar{q}}{n}, \qquad \sigma_{e_n,\mathrm{stat}} = \frac{\sigma_{\bar{q},\mathrm{stat}}}{n}. \tag{25}$$

These standard errors describe the scatter of the measurements within each group. They do not include shared systematic uncertainties, such as uncertainty in the plate separation or voltage. Any such contributions should be reported separately and should not be reduced by $1/\sqrt{N}$.

If the level estimates are independent and only their statistical uncertainties are combined, the inverse-variance weighted estimate is

$$e_{\mathrm{weighted}} = \frac{\sum_n e_n/\sigma^2_{e_n,\mathrm{stat}}}{\sum_n 1/\sigma^2_{e_n,\mathrm{stat}}}, \qquad \sigma_{e,\mathrm{weighted}} = \left(\sum_n \frac{1}{\sigma^2_{e_n,\mathrm{stat}}}\right)^{-1/2}. \tag{26}$$

The percentage difference from the accepted elementary charge is

$$\text{percentage difference} = \frac{|e_{\mathrm{measured}} - e_{\mathrm{accepted}}|}{e_{\mathrm{accepted}}} \times 100\%. \tag{27}$$

6.1 Uncertainty on e_1 from the expected A.

Using the expected value of A, the uncertainty in e_1 can be propagated from Eq. 23,

$$\sigma_{e_1} = 1.7 \pm 0.1\mathrm{C}.$$

6.2 Uncertainty on e_1 from the fitted A.

The Cunningham fit is $e_{\mathrm{app}}^{2/3} = c + m\lambda/r$ with $c = e_1^{2/3}$ and $A = m/c$. Its parameters and covariance come from the least-squares Jacobian, scaled by the residual variance. This gives $c = (1.3506 \times 10^{-13} \pm 2.63 \times 10^{-12})\,\mathrm{C}^{2/3}$, $m = (0.936 \pm 458.753) \times 10^{-13}\,\mathrm{C}^{2/3}$, and $A = 0.6927 \pm 0.55$. Since $e_1 = c^{3/2}$, propagating u_c gives

$$\sigma_{e_1} = \frac{3}{2}\, c^{1/2}\, u_c = \frac{3}{2}\, e_1 \frac{u_c}{c}, \tag{28}$$

so

$$e_1 = (4.9633 \times 10^{-20} \pm 1.45 \times 10^{-18})\,\mathrm{C}, \tag{29}$$

which is 69.0% below the accepted value, but is not a useful estimate because its uncertainty is much larger than its central value. With the analytic A the unweighted fit gives $e_1 = (1.72 \pm 0.01) \times 10^{-19}\,\mathrm{C}$, but that error has no contribution from A, b or d, so it is optimistic. As above, the quoted uncertainties exclude the exact-treated constants.

7 Discussion

7.1 Quantisation of charge

The central result is that the apparent charges do not form a continuum. The 28 drops fall into three well separated groups, with weighted means $q_1 \approx 1.77$, $q_2 \approx 3.44$ and $q_3 \approx 5.39$ in units of 10^{-19} C. These stand in the ratio $1 : 1.94 : 3.05$, close to $1 : 2 : 3$, so each drop carries a small whole number of a single fundamental unit. A fit of $q = en + c$ is consistent with a zero intercept, as expected if no charge is carried by a drop with $n = 0$. No drop was found between the groups, even though the oil droplets were charged by friction in an uncontrolled way, so a continuous charge distribution would populate the gaps. This is the key conclusion of Millikan's experiment: charge is quantised, and the value of the unit e can be read from the spacing of the levels.

The experiment supports this, but does not prove it beyond doubt. The level estimates have $\chi^2 = 9.34$ for two degrees of freedom, so the spread between groups is larger than the within-group standard errors imply. The likely cause is uncertainty not in the error budget (drop radius, viscosity, a small error in the assigned n for individual drops), rather than a failure of quantisation. Only three levels were reached, and the $n = 3$ group has few drops with large individual uncertainties.

7.2 Brownian motion and the slip correction

The drops have radii of order 10^{-6} m, comparable with the mean free path of air, $\lambda \approx 7 \times 10^{-8}$ m, so $\lambda/r \approx 0.07$–0.11. The air therefore cannot be treated as a continuous fluid: the drop is bombarded by individual molecules, which is the same molecular bombardment that produces Brownian motion. Two consequences appear in the data. First, the Cunningham slip correction is expected from the molecular nature of air, and applying the analytic correction shifts the three level estimates. However, the fitted slope is too uncertain here to measure this effect independently. Second, the random molecular impacts make the measured fall and rise times fluctuate from one passage to the next, which contributes to the scatter of the speeds and hence of the charges. The experiment is therefore an exploration of Brownian motion as well as of charge quantisation: the slip correction tests the kinetic picture of a gas, and the transit-time scatter is the Brownian noise on the terminal velocity.

7.3 Success of the experiment and Avogadro's number

The experiment was successful in its main aims. It shows clearly grouped charges and gives $e_1 = (1.7 \pm 0.1) \times 10^{-19}$ C if the analytic A is used, which is 7.1% above the accepted value of 1.602×10^{-19} C. The free fit gives $A = 0.6927 \pm 0.55$, consistent with the analytic value within 0.9σ. Its central value corresponds to $n = 1.4236 \times 10^{25}\,\mathrm{m}^{-3}$ and Avogadro's number (Appendix, Section on Avogadro's Number),

$$N_\mathrm{A} = (3.51 \pm 2.79) \times 10^{23}\,\mathrm{mol}^{-1},$$

whose central value is 41.7% below $6.022 \times 10^{23}\,\mathrm{mol}^{-1}$, but consistent within 0.9σ. Thus, the experiment estimates the charge of the electron from the grouped drop charges and

gives a rough independent estimate of Avogadro's number. This is the same link made by Einstein's and Perrin's analysis of Brownian motion [2] [7], where $N_A = R/k_B$ is fixed by the thermal motion of small particles. Using the analytic A returns N_A exactly, but that is circular, so only the fitted-A value is an independent result.

7.4 Limitations

The measured values of e_1 are high. The weighted level estimate is 10.8% above the accepted elementary charge, and the origin-constrained fit is 9.8% high. The propagated uncertainties treat plate spacing, viscosity, oil density, the slip constant and the molecular diameter as exact, so they do not capture calibration uncertainty in these quantities. A shared scale error would affect all drops alike; since $q \propto \eta^{3/2}$, a few-percent error in η (which varies with temperature) would shift e_1 by more than that. The data does not identify which systematic effect, if any, explains the difference. The calculation also assumes spherical drops, terminal motion, a uniform field between the plates and the stated Cunningham correction; departures from these are not in the reading uncertainties. The low reduced chi-squared of the individual-point fits should be read in light of the assumed reading resolutions and omitted calibration terms, not as independent confirmation of the uncertainty model. The N_A uncertainty comes only from A_{fit} and omits d and b, on which it depends strongly ($\propto 1/d^2 b$).

8 Conclusion

The analytic Cunningham coefficient is $A = \sqrt{2}\pi d^2 b/(k_{\mathrm{B}}T) = 1.188$. Millikan's method of fitting $q_{\mathrm{app}}^{2/3}$ against λ/r gives $A = 0.6927 \pm 0.55$, consistent with the analytic value within 0.9σ. Correcting the apparent charges with the analytic A gives $e_1 = (1.7 \pm 0.1) \times 10^{-19}\,\mathrm{C}$, which is 7.1% above the accepted value. The free fit is too uncertain to give a useful alternative estimate. This result supports charge grouping, while a full uncertainty budget must also include uncertainty in b, the molecular diameter and shared apparatus constants.

9 References

References

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A Appendix

A.1 Cunningham correction factor

The Cunningham correction factor accounts for the slip of air molecules around small particles, which affects the drag force experienced by the oil drops. A general empirical form is

$$C_{\mathrm{c}} = 1 + \frac{\lambda}{r}\left(\alpha + \beta \exp\left(-\frac{\gamma r}{\lambda}\right)\right), \tag{30}$$

where λ is the mean free path of air molecules, r is the drop radius, and α, β, and γ are empirical constants. For the calculations in this report, the Leybold apparatus approximation is used:

$$C_{\mathrm{c}}(r,p) = 1 + \frac{b}{rp} = 1 + \frac{A}{r}, \qquad A = \frac{b}{p}, \tag{31}$$

with $b = 80\,\mu\mathrm{m\,hPa}$ for the apparatus and air pressure p. This is the correction used in the drag and charge calculations in the main text. Here $b/p = A\lambda$, so $A = b/(p\lambda)$ is the dimensionless coefficient used in the main text, and λ/r is the Knudsen-type ratio. The linear form is the first-order slip expansion; for $\lambda/r = 0.0674$–0.1163 in this experiment the exponential term in the general form is negligible. The coefficient b depends on gas–surface accommodation and is an empirical, apparatus-quoted constant (Leybold $b = 8.0 \times 10^{-3}\,\mathrm{Pa\,m}$ versus the slightly different value quoted in the Oxford notes), so it cannot be derived exactly from the Knudsen number alone. The general form has the empirical coefficients α, β, γ in place of the single A.

A.2 Derivation of the fit model and analytic A

Terminal-velocity motion with slip gives a Stokes drag reduced by $1 + A\lambda/r$, so the apparent radius r_0, computed without slip, obeys

$$r_0^2 = r^2 + A\lambda r \;\Rightarrow\; r = \sqrt{r_0^2 + \tfrac{(A\lambda)^2}{4}} - \tfrac{A\lambda}{2}, \tag{32}$$

and the charge scales as $q \propto r^{3/2}$ at fixed velocity ratio, giving $q_{\mathrm{app}}^{2/3} = q^{2/3}(1 + A\lambda/r)$ as used in the Analysis. The fit minimises the residuals of this relation over all 28 drops with one A and one $e_1^{2/3}$. Using $\lambda = k_{\mathrm{B}}T/(\sqrt{2}\pi d^2 p)$,

$$A = \frac{b}{p\lambda} = \frac{\sqrt{2}\pi d^2 b}{k_{\mathrm{B}}T}, \tag{33}$$

so the pressure cancels. The uncertainty of the free fit was estimated from the least-squares covariance scaled by the residual variance.

A.3 Mean Free Path of Air Molecules

The mean free path of air molecules, denoted by λ, is the average distance a molecule travels between collisions. It is given by the expression:

$$\lambda = \frac{k_{\mathrm{B}} T}{\sqrt{2}\pi d^2 p}, \tag{34}$$

where k_{B} is the Boltzmann constant, T is the absolute temperature, d is the effective diameter of an air molecule, and p is the air pressure. This parameter is crucial for calculating the Cunningham correction factor for small oil drops.

Using $T = 296.55\,\mathrm{K}$, $p = 1000\,\mathrm{hPa}$, and an effective air-molecule diameter $d = 0.370\,\mathrm{nm}$ gives the mean free path

$$\lambda = 6.7315 \times 10^{-8}\,\mathrm{m}.$$

A.4 Calculating Avogadro's Number

For molecules of diameter d, the collision cross-section is πd^2. In a gas with number density n, the mean free path, neglecting the relative motion of molecules, is

$$\lambda_0 = \frac{1}{n\pi d^2}.$$

Accounting for the relative motion introduces a factor of $\sqrt{2}$, giving

$$\lambda = \frac{1}{\sqrt{2}\, n\pi d^2},$$

consistent with Eq. 34, where $n = p/(k_{\mathrm{B}} T)$.

Avogadro's number is related to the number density by

$$N_{\mathrm{A}} = n V_{\mathrm{molar}},$$

where $V_{\mathrm{molar}} = RT/p$ is the molar volume of an ideal gas. Using the slip-correction constant $A = b/(p\lambda)$, we obtain

$$\lambda = \frac{b}{pA}, \qquad n = \frac{pA}{\sqrt{2}\,\pi d^2 b}.$$

Substituting this expression into the equation for N_{A} eliminates the pressure, yielding

$$\boxed{N_{\mathrm{A}} = \frac{ART}{\sqrt{2}\,\pi d^2 b}}.$$

Using $R = 8.3145\,\mathrm{J \cdot mol^{-1} \cdot K^{-1}}$, $T = 296.55\,\mathrm{K}$, $d = 0.370\,\mathrm{nm}$, and $b = 8.0 \times 10^{-3}\,\mathrm{Pa \cdot m}$, together with the fitted value $A = 0.6927 \pm 0.55$, gives

$$\lambda = 1.1549 \times 10^{-7}\,\mathrm{m}, \qquad n = 1.4236 \times 10^{25}\,\mathrm{m^{-3}}.$$

The resulting estimate of Avogadro's number is

$$\boxed{N_{\mathrm{A}} = (3.51 \pm 2.79) \times 10^{23}\,\mathrm{mol}^{-1}}.$$

The central value is 41.7% below the accepted value of $6.022 \times 10^{23}\,\mathrm{mol}^{-1}$ [7], but remains consistent with it within 0.9σ.

The quoted uncertainty accounts only for the uncertainty in A_{fit}; it excludes uncertainties in d and b, on which the estimate depends strongly, with $N_{\mathrm{A}} \propto 1/(d^2 b)$ for fixed A.

For comparison, substituting the analytic value $A = 1.188$ reproduces the accepted value of N_{A}. However, this agreement is circular rather than an independent measurement: since $A \propto d^2 b/(k_{\mathrm{B}} T)$, substitution reduces the expression to $N_{\mathrm{A}} = R/k_{\mathrm{B}}$. The result therefore serves only as a consistency check, not an independent determination of Avogadro's number [5].